Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | Efficient and Scalable AI Deployment through Software Infrastructure and Transfer Learning | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√ ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | AI infrastructure and deployment support | | Knowledge Services | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Software | Software solutions | R&D / Engineering services | Other R&D / Engineering-related Technologies |
| Concept of Proposed Technology | AI technologies are increasingly embedded in industrial products and services, yet many SMEs face significant barriers in deploying, scaling, and maintaining AI systems. While specialized AI hardware (e.g. accelerators and edge devices) can deliver substantial performance and energy-efficiency gains, these benefits are often undermined by fragmented software stacks, hardware-specific tooling, and | | | |

complex deployment workflows.

The proposed technology enables efficient and scalable AI deployment through software infrastructure and transfer learning. It reduces deployment and learning effort by decoupling AI models from hardware-specific environments via software-level abstractions, while supporting reuse of learned representations across related systems and tasks.

Key functions of the technology include:

- Support for portable and maintainable AI deployment across heterogeneous hardware environments
- Reduction of manual configuration and integration effort through software infrastructure abstractions
- Reuse of learned representations via transfer learning to reduce training time, data requirements, and computational cost

The technology is designed as supporting infrastructure rather than a proprietary platform. It integrates with existing AI pipelines and enables scalable deployment, allowing SMEs to reuse models, workflows, and deployment concepts across products, systems, and markets.

This work builds on TNO's prior research in adaptive data-processing infrastructures and distributed AI learning, focusing on practical constraints and architectural trade-offs rather than vendor-specific solutions. Earlier research on adaptive, on-the-fly reconfiguration of distributed processing pipelines demonstrated how AI workflows can evolve at runtime without full redeployment (1). More recent studies explored federated and transfer learning in industrial contexts, including geothermal energy systems (2), decentralized secure modeling (3), and low-voltage grid state estimation (4), highlighting both the potential and limitations of distributed learning under real operational and data-governance conditions.

In parallel, TNO has contributed to the development of open and reusable AI assets, including open-source models, datasets, and reference implementations. Examples include the *BioCube* [1] multimodal dataset for biodiversity research (5) and the *BioAnalyst* foundation model (6). Designed as lightweight, adaptable building blocks, these assets demonstrate how AI models and infrastructures can support reuse across multiple downstream applications while remaining transparent and accessible.

This experience has been further solidified by applied European and national projects such as

[1] BioDT/BioCube · Datasets at Hugging Face

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| | ENERSHARE [2](trustworthy energy data spaces and federated data sharing), ZEFES [3](zero-emission freight transport with complex operational constraints), and FLEXMCS [4](scalable, energy-efficient megawatt charging infrastructure for electric heavy-duty vehicles). In these projects, AI components were embedded within broader software and system architectures, reinforcing the need for deployment approaches that are scalable, maintainable, and independent of proprietary cloud or platform lock-in.<br><br>As an independent research and technology organization, TNO does not act as a commercial AI platform provider. This ensures the proposed technology remains vendor-neutral and transparent, enabling SMEs to adopt and adapt the resulting infrastructure within their own environments. The R&D therefore emphasizes feasibility, architectural choices, and validation in close collaboration with industrial partners. |
| **Rationale for Joint R&D Project** | Many SMEs face persistent challenges in deploying and scaling AI systems in operational environments. These challenges extend beyond model development and stem primarily from tight coupling between AI software, hardware platforms, and learning workflows. As a result, SMEs encounter high integration effort, limited hardware portability, and difficulties maintaining AI systems over time. Existing deployment solutions are often tailored to large enterprises or specific vendors, limiting their suitability for SME contexts. In addition, AI models are frequently retrained from scratch for each deployment, even when systems share similar characteristics, leading to unnecessary effort, higher computational cost, and slower scaling.<br><br>Addressing these issues requires targeted R&D that jointly considers software infrastructure design and learning reuse through transfer and federated learning. This is not a purely algorithmic problem but a system-level challenge involving architectural choices, deployment trade-offs, and validation under realistic operational constraints. Most SMEs lack the resources and interdisciplinary expertise needed to develop and validate such integrated solutions independently, as the required capabilities span AI modeling, software engineering, and hardware-aware deployment.<br><br>TNO brings expertise in AI software architecture, adaptive data-processing infrastructures, and distributed learning, developed through prior industrial and applied research. However, validating feasibility, scalability, and operational relevance requires close collaboration with industrial partners operating AI systems on real hardware and within real business constraints.<br><br>Korean SMEs provide access to diverse hardware environments, deployment practices, and |

[2] https://enershare.eu/

[3] ZEFES | Taking Zero-Emission Long-Haul Freight Transport in Europe to the next level

[4] FLEX MCS

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| | application domains that complement European contexts. Joint R&D within the TIPA framework enables early validation across heterogeneous settings, supporting the development of software infrastructure and learning approaches that are robust, portable, and applicable across international markets.<br><br>Through this collaboration, the project aims to reduce technical and adoption risks for SMEs while establishing a foundation for scalable, maintainable, and resource-efficient AI deployment strategies that can be integrated into future products and services. |
| **Key Words** | 1. Efficient AI Deployment<br>2. Software Infrastructure<br>3. Transfer Learning<br>4. AI Scalability<br>5. Resource-Efficient AI |

## 3. Research and Development Objectives & Plan

| **Technology Readiness Level** (TRL)[2] | **Start of the Project** | Level 3 | **End of the Project** | **End of Phase 1: TRL 4**<br>**Target TRL at End of Phase 2: TRL 5–6** |
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| **R&D Objectives** | **Phase 1 Objectives (Month 1–6)**<br>– Assess the feasibility of software infrastructure abstractions for portable AI deployment across heterogeneous hardware environments<br>– Evaluate the potential of transfer and federated learning for reuse of learned representations in SME-relevant settings<br>– Identify architectural options, technical constraints, and design trade-offs<br>– Define a focused, realistic scope and success criteria for a potential Phase 2<br>**Phase 2 Objectives (Month 7–36):** Building on Phase-1 results, Phase 2 aims to:<br>– Implement and validate selected deployment and learning approaches in operational SME environments<br>– Demonstrate scalability, robustness, and maintainability across multiple deployments or system configurations<br>– Support integration of the developed approaches into SME products, services, or internal workflows | | | |
| **R&D Target Specifications** | **Phase 1 – Targets** | | | |

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| | – **Portable AI deployment feasibility**: Demonstrate deployment of a representative AI workload across at least two heterogeneous execution environments without model redesign, using prototype-level tooling.<br>– **Reduced deployment effort**: Provide qualitative or semi-quantitative evidence of reduced configuration and integration effort compared to a baseline SME workflow (indicatively ~20–30%, depending on system complexity).<br>– **Learning reuse feasibility**: Demonstrate reuse of learned representations across related systems or tasks using transfer and/or federated learning, with indicative reductions in training time or data requirements (~20–30%) in a controlled setting.<br>– **Prototype-level integration**: Demonstrate a prototype integration of deployment infrastructure and learning reuse within an existing AI pipeline for one representative SME application.<br>**Phase 2 – Targets:** Phase 2 targets focus on validation and scaling:<br>– Operationally validated AI deployment infrastructure in pilot or operational SME environments<br>– Measurable reductions in deployment effort, retraining time, or operational overhead<br>– Reusable reference architectures, deployment guidelines, and design patterns supporting broader SME adoption |
| **R&D Plan** | The project is structured in two phases, **Phase 1 (6 months)** and **Phase 2 (approximately 30 months).** This phased approach ensures early insight with limited risk, followed by focused development once feasibility and value have been demonstrated.<br>**Phase 1 – Feasibility and Design Definition (Month 1–6)**<br>Establish a shared understanding of the use case, assess feasibility of key concepts, and define a realistic and focused scope for Phase 2.<br>**Key Tasks and Timeline**<br>– **Task 1 – Partner alignment and baseline definition (Month 1–2):** Define the representative deployment scenario and document baseline AI deployment and learning workflows, including hardware, data, and operational constraints.<br>– **Task 2 – Exploratory design of deployment and learning concepts (Month 2–3):** Identify and compare candidate software infrastructure abstractions and transfer/federated learning approaches; define architectural options and design trade-offs.<br>– **Task 3 – Prototype-level feasibility exploration (Month 3–5):** Implement limited prototype components and perform exploratory evaluation in a controlled or representative setting.<br>– **Task 4 – Consolidation and Phase-2 scoping (Month 5–6): Consolidate** results, assess |

scalability and organizational challenges, and define objectives, success criteria, and resource needs for Phase 2.

**Milestones (Phase 1)**

**M1:** Partner alignment completed and baseline documented

**M2:** Feasibility of deployment and learning concepts evaluated

**M3:** Evaluation report and Phase-2 roadmap delivered

**Phase 2 – Development, Validation, and Scaling (Month 7–36)**

Develop, validate, and demonstrate scalable AI deployment infrastructure and learning reuse in operational SME environments.

**Key Tasks and Timeline**

**Task 5 – Implementation of scalable deployment infrastructure (Month 7–14)**
Implement and refine software infrastructure components enabling portable and maintainable AI deployment across heterogeneous hardware environments, based on Phase-1 architectural choices.

**Task 6 – Validation of learning reuse approaches (Month 12–20)**
Implement and validate transfer and/or federated learning approaches across multiple systems, tasks, or deployments, assessing learning reuse under realistic operational constraints.

**Task 7 – Integration into SME workflows (Month 18–26)**
Integrate deployment and learning components into SME workflows, including CI/CD pipelines, monitoring, and operational processes where relevant.

**Task 8 – Performance and scalability evaluation (Month 24–32)**
Evaluate performance, robustness, and scalability of the integrated solution under realistic operational conditions.

**Task 9 – Consolidation, valorisation, and guidance (Month 30–36)**
Consolidate results into reusable components, reference architectures, and design guidelines, and support valorisation and adoption activities aligned with SME business models.

**Milestones (Phase 2)**

**M4:** Core deployment infrastructure implemented and integrated in SME environment

**M5:** Learning reuse validated across multiple systems or deployments

**M6:** Performance, scalability, and operational impact evaluated

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| | **M7:** Reference architectures, guidelines, and exploitation inputs delivered<br><br>Phase 2 will be scoped and executed based on Phase-1 findings, ensuring technical grounding and alignment with SME needs. |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | TNO will act as the **technical and methodological lead**, responsible for the design, feasibility assessment, and validation of software infrastructure and learning concepts in close collaboration with the SME partner.<br><br>**In Phase 1, TNO will:**<br><br>– Design and assess software infrastructure and architectural concepts supporting efficient and scalable AI deployment across heterogeneous hardware environments, tailored to the SME's application context.<br><br>– Identify, select, and evaluate transfer and federated learning approaches relevant to the use case, focusing on feasibility, integration effort, and potential reductions in training and deployment cost.<br><br>– Perform system-level analysis to identify technical assumptions, constraints, and design trade-offs related to deployment, learning reuse, and scalability.<br><br>– Consolidate findings into clear design recommendations, feasibility conclusions, and priorities to support decision-making and define a realistic scope for Phase 2.<br><br>**In Phase 2, TNO will:**<br><br>– Support the implementation, validation, and refinement of selected deployment and learning approaches based on Phase-1 outcomes.<br><br>– Contribute to performance, robustness, and scalability evaluation in operational or pilot SME environments.<br><br>– Support the development of reusable reference architectures, design guidelines, and technical documentation to facilitate adoption and scaling. |
| **Expected R&R of Korean SME Partner** | The Korean SME will act as the **industrial application and validation partner**, ensuring that the project remains grounded in real operational, technical, and business needs.<br><br>**In Phase 1, the Korean SME will:**<br><br>– Provide a representative AI application context, including relevant information on operational |

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| | workflows, data characteristics, hardware environment, and deployment constraints.<br>– Participate actively in feasibility validation activities, including review of intermediate results and feedback on practicality, integration effort, and operational relevance.<br>– Contribute to the joint assessment of business relevance and valorisation potential, helping to identify success criteria and priorities for a potential Phase 2.<br>**In Phase 2, the Korean SME will:**<br>– Support pilot deployments and operational validation of the developed approaches within relevant systems or environments.<br>– Contribute to evaluation of deployment effort, learning reuse, and operational impact under realistic conditions.<br>– Explore pathways for integration of the project outcomes into products, services, or internal workflows, supporting future commercialization or internal adoption. |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | **Expected Outcomes – Phase 1 (Feasibility Phase)**<br>At the end of Phase 1, the project is expected to deliver:<br>– A **feasibility-validated concept** for software infrastructure supporting efficient and scalable AI deployment across heterogeneous hardware environments.<br>– A **validated indication of learning reuse potential** through transfer and/or federated learning, explored in a controlled or representative SME context.<br>– **Indicative evidence of reduced deployment and training effort**, compared to baseline SME workflows, based on exploratory evaluation rather than full operational deployment.<br>– **Architectural design guidance and recommendations**, including identified technical constraints, assumptions, risks, and priorities for further development.<br>These outcomes provide the technical, organizational, and strategic insight required to make an informed decision on whether and how to proceed to Phase 2.<br>**Expected Outcomes – Phase 2 (Development and Scaling Phase)**<br>Building on the results of Phase 1, Phase 2 is expected to deliver:<br>– **Implemented and validated software infrastructure components** enabling portable, maintainable, and efficient AI deployment in real SME environments. |

- **Operational validation of transfer and/or federated learning approaches**, demonstrating reuse of learned representations across systems, deployments, or tasks.
- **Measured reductions in deployment effort, retraining time, and/or operational overhead**, compared to conventional AI deployment practices.
- **Reusable reference architectures, design patterns, and technical guidelines** supporting adoption across SME products, services, or markets.

These outcomes are intended to support both technical scaling and practical integration into SME offerings or internal operations.

**Impact**

**For SMEs, the project enables:**

- Reduced deployment risk and integration effort by decoupling AI models from hardware-specific implementations.
- Improved reuse of AI investments through transfer and federated learning, lowering retraining effort, data requirements, and computational cost.
- Increased predictability and maintainability of AI-enabled systems, simplifying updates, adaptation, and long-term support.

**From a business perspective, these improvements support:**

- Faster time-to-market for AI-enabled products and services.
- Lower engineering and operational costs per deployment.
- Clearer pathways from pilot projects to scalable offerings, reducing the risk of stalled innovation.

**From an international collaboration perspective:**

- Korean SMEs benefit from early validation of scalable AI deployment approaches with a European research partner.
- European partners gain exposure to Korea's advanced hardware ecosystems and deployment practices.
- Jointly developed approaches are validated across heterogeneous environments, strengthening their applicability in international markets.

Overall, the project contributes to more sustainable and scalable adoption of AI technologies by SMEs, strengthening competitiveness while reducing unnecessary technical and operational overhead.

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